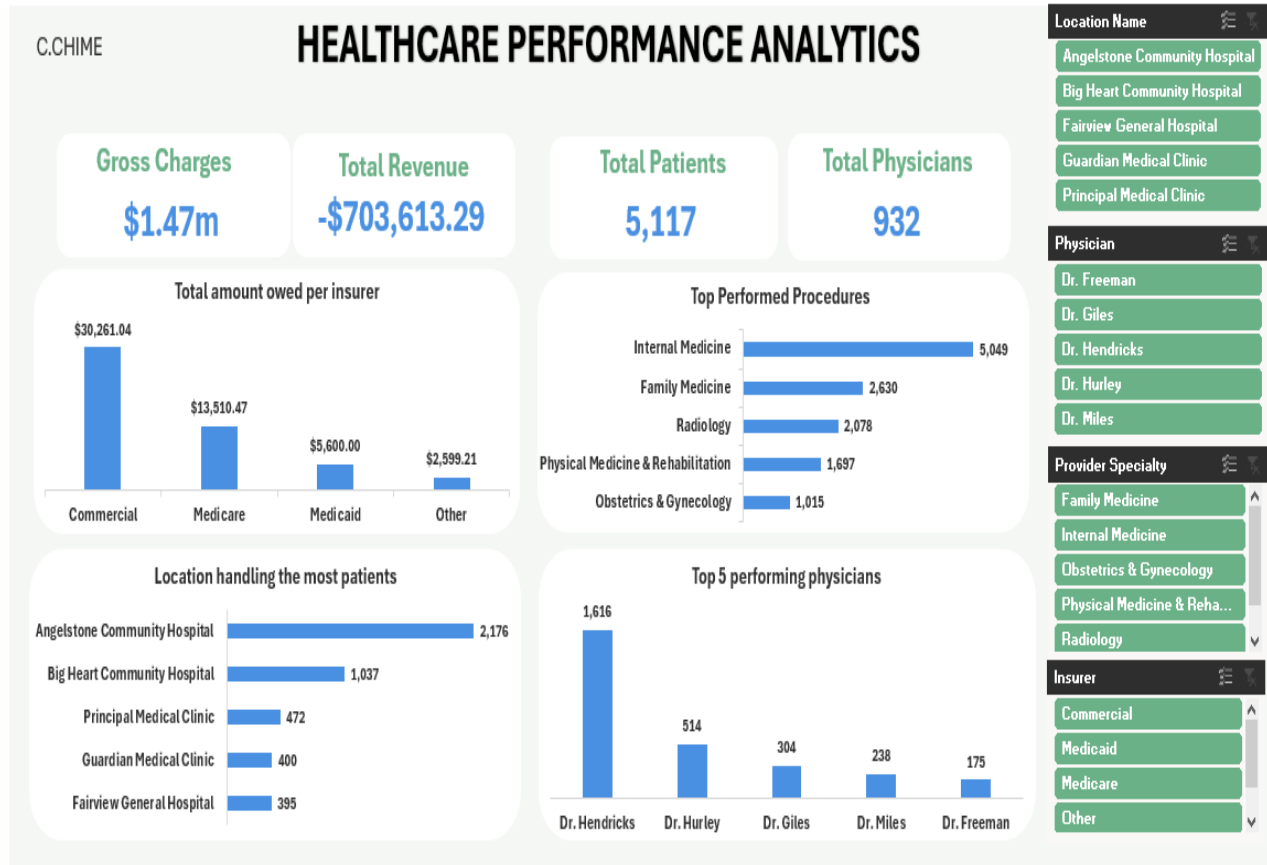


HEALTHCARE PERFORMANCE ANALYTICS

ABSTRACT



[View Excel Dashboard](#)

The negative revenue value reflects the accounting convention used in the dataset, where payments and adjustments are stored as negative values to represent reductions against gross charges rather than actual financial losses

INTRODUCTION

Background

This study focuses on understanding how healthcare data can be transformed into meaningful insights that support better clinical decisions, operational efficiency, and financial sustainability within a hospital environment. This study demonstrates how data analytics can bridge the gap between raw hospital records and informed decision-making.

Problem Statement

The core problem addressed in this analysis is the difficulty faced by healthcare organizations in monitoring performance across patients, physicians, procedures and revenue streams simultaneously.

Research Questions

1. Which insurer contributes most to outstanding receivables?
2. Who are the highest performing physicians?
3. Which procedure drives hospital operation?
4. Which hospital location handles most patients?

Objectives

1. To identify which insurer payer has the most outstanding debt.
2. To determine the highest performing physicians.
3. To ascertain the top performed procedures.
4. To find out the hospital location that handles the most patient.

DATA DESCRIPTION

Data Sources: The data used for this analysis comes from a healthcare data warehouse structured as a star-schema consisting of multiple related tables stored in CSV format.

Data Characteristics: The dataset is relational, structured, and analytical in nature. It consists of a central fact table that records healthcare encounters and financial measures, and multiple dimension tables describing patients, physicians, locations, dates, procedures and transactions. It contains both categorical attributes (such as gender, physician specialty, procedure type, and facility location) and numerical measures (such as charges, payments, costs, and length of stay)

METHODOLOGY

Data Cleaning

1. Duplicate Removal: In my data cleaning process, I checked for duplicates by counting the unique patient ID (primary key) using SQL queries.

2. Standardizing state format: While exploring my dataset, I discovered that the state column in the patient table was abbreviated so I had to; distinct count each state, get the unique state, and standardize each state to its full-name.

Exploratory Data Analysis

Financial Performance Analysis

In my analysis, I discovered that the total revenue had a negative value, I then engaged in a full financial analysis and discovered the following;

Gross Charges (\$1,472,266.81): This is the total amount billed by the hospital to the patients before insurance negotiations.

Payments (\$-703,613.29): This is the money that was actually received from insurers or patients and this represents the true revenue received by the hospital.

Adjustments (\$-716,682.80): This is the amount of write-offs and contractual discounts from the gross charge. This shows a high adjustment rate that is almost half of total charges were written off.

Accounts Receivables (\$51,970.72): This is the outstanding unpaid balance from the insurers.

Key Performance Indicators

a. Gross Charge: The total amount billed for all services performed, before any payments, adjustments, or write-offs are applied.

b. Total Revenue: The net amount the hospital actually earns, calculated as payments received minus adjustments applied.

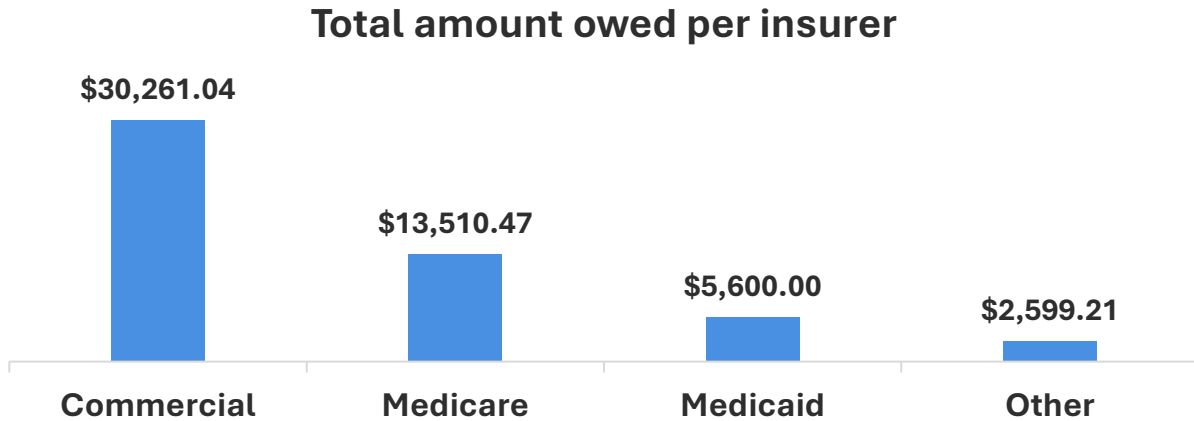
c. Total Patients: The distinct count of unique patients who received at least one service within the dataset.

d. Total Physicians: The unique count of physicians who provided at least one service in any of the healthcare's location.

ANALYSIS AND FINDINGS

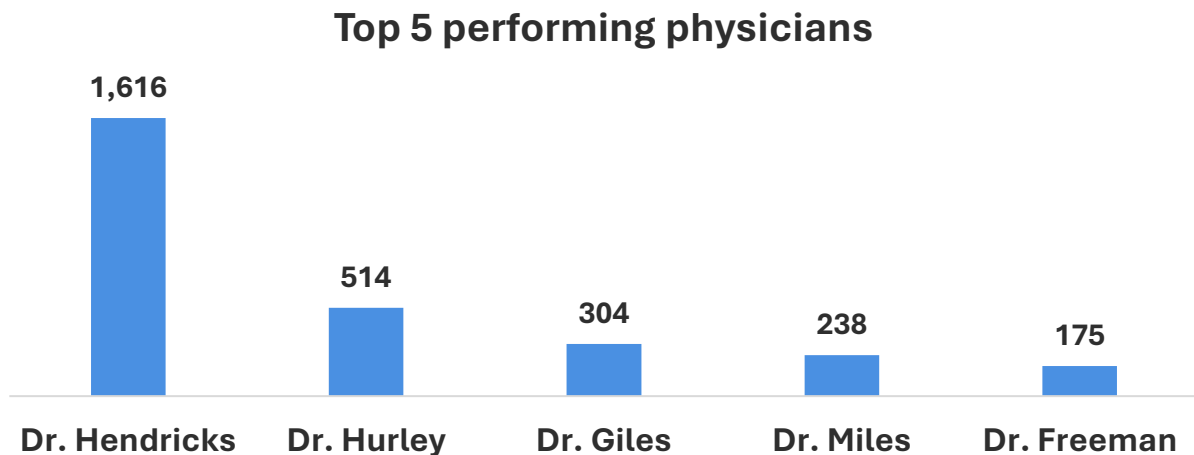
Analysis Questions

1. Which insurer contributes most to outstanding receivables?



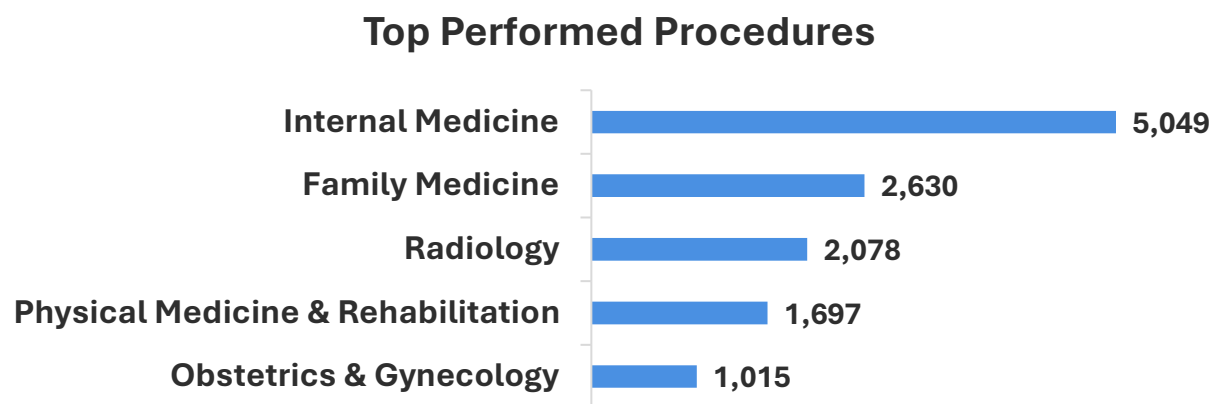
The chart above reveals that commercial insurance contributes the highest amount to outstanding receivables with approximately \$30,251.04, compared to Medicare, Medicaid and other insurers. This indicates that commercial insurers account for the largest share of unpaid claims and delayed reimbursements.

2. Who are the highest performing physicians?



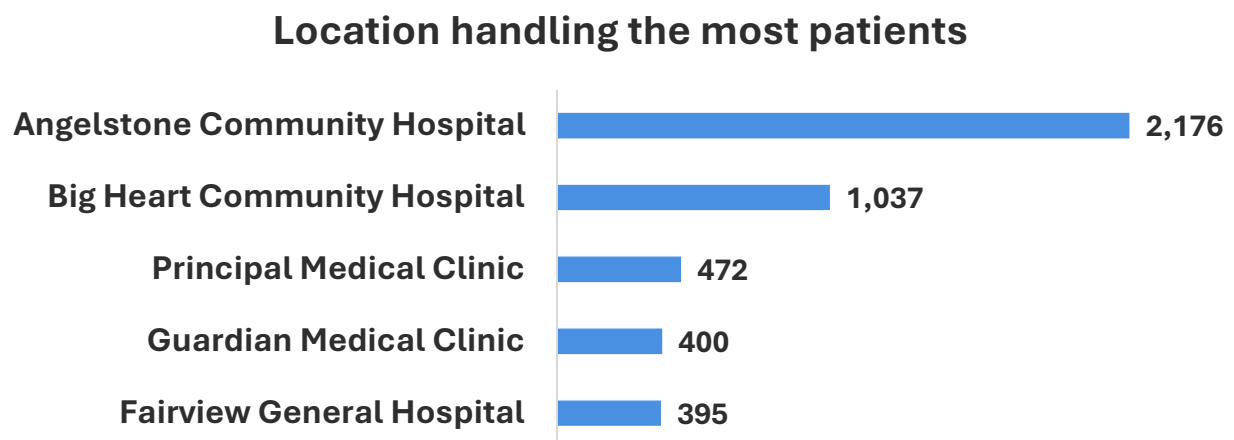
Looking at the physician chart, there is a clear indication that Dr. Hendricks is the highest performing physician with 1,616 recorded procedures or patient interactions, far ahead of other physicians; Dr. Hurley, Dr. Giles, Dr. Miles, and Dr. Freeman. The significant performance gap between Dr. Hendricks and the remaining physicians suggests a concentration of patient demand and workload around a single provider.

3. Which treatment drives hospital operation?



The treatment and procedure analysis identifies internal medicine as the top performed specialty driving hospital operation with 5,049 procedures performed. Family medicine follows with 2,630 procedures while radiography, physical medicine and rehabilitation, and Obstetrics and Gynecology contribute lower volumes.

4. Which hospital location handles most patients?

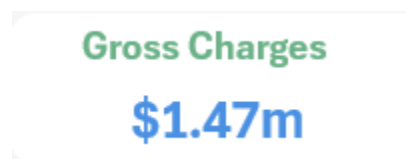


The hospital analysis chart shows that Anglestone Community Hospital handles the highest number of patients with 2,176 patient visits, more than double the patient volume managed by Big Heart Community Hospital with 1,037 patients. Principal Medical Clinic, Guadian Medical Clinic, and Fairview General hospital recorded lower patient counts.

Key Performance Indicators

For my analysis, I had the following Key Performance Indicators

a. Gross Charge:



This KPI shows the total amount charged by the hospital for services rendered before deductions, adjustments and insurance write-offs are applied.

b. Total Revenue:



This represents the actual financial outcome after adjustments reductions and receivables are accounted for. The negative sign indicates that the organization experienced more deductions and unpaid claims and this indicates a financial pressure despite the substantial gross charge.

c. Total Patients:



This measures the total number of patients served across all hospitals and clinics within the reporting period. The value indicates a strong patient volume and demonstrates that the organization is actively providing healthcare services to a large population.

d. Total Physicians:

Total Physicians

932

This represents the total number of healthcare service providers involved in delivering medical services across the organization. A physician count of 92 reflects a large healthcare workforce supporting patient care and hospital operation.

RESULT

Summary of Findings

The analysis shows that commercial insurance has the highest outstanding receivables, making it the biggest contributor to unpaid balances. Dr. Hendricks is the top performing physician with the highest number of procedures, while Internal medicine is the leading specialty driving hospital activity. Angelstone community records the highest patient volume. Despite strong operational output of \$1.47M in gross charges, the organization reports a negative revenue of \$-703,613.29, indicating financial inefficiencies.

Visualizations

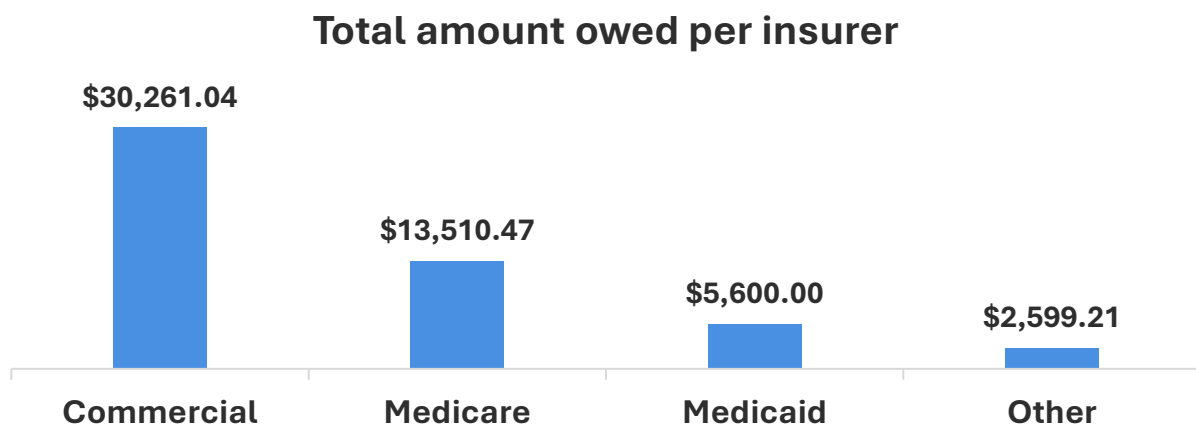


Figure 1: Rank of total amount owed

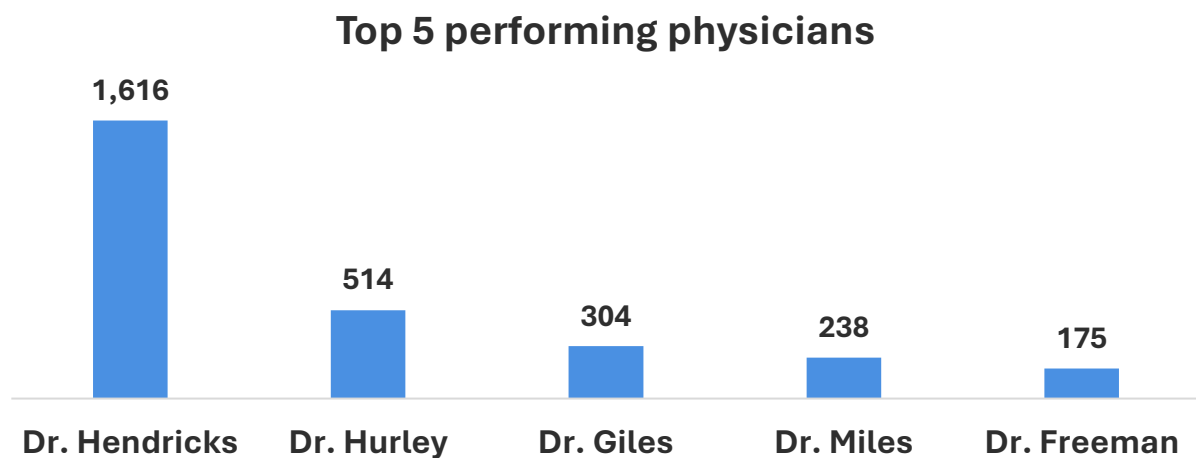


Figure 2: Top Performing Physicians

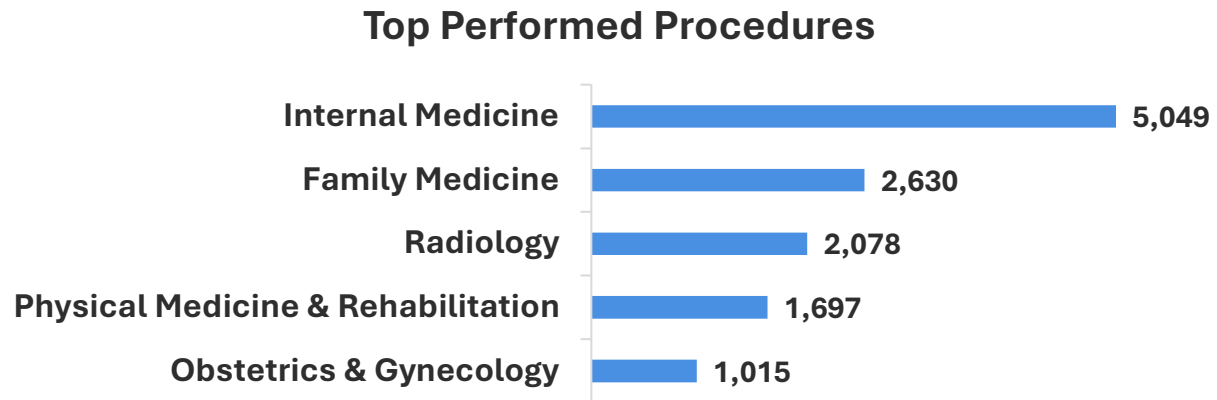


Figure 3: Top performed procedures

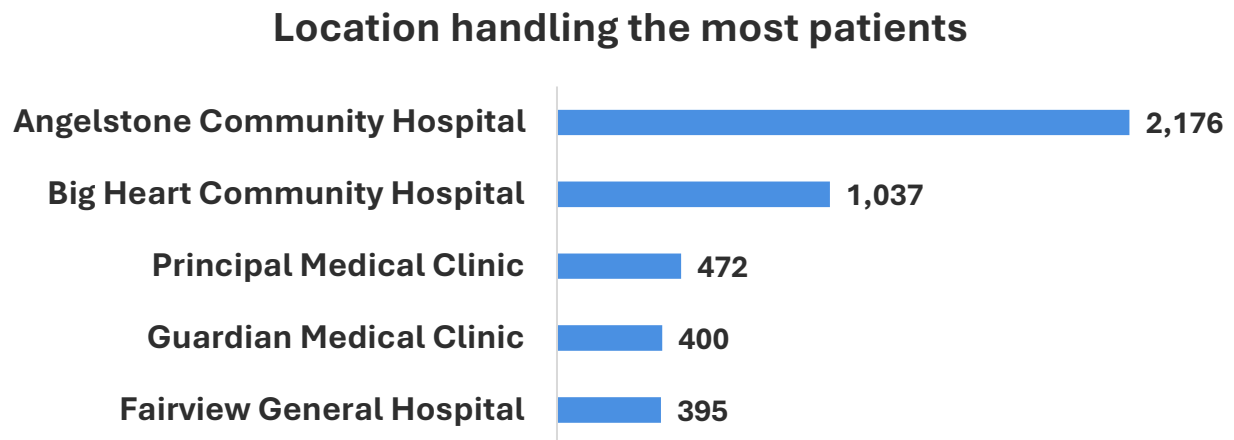
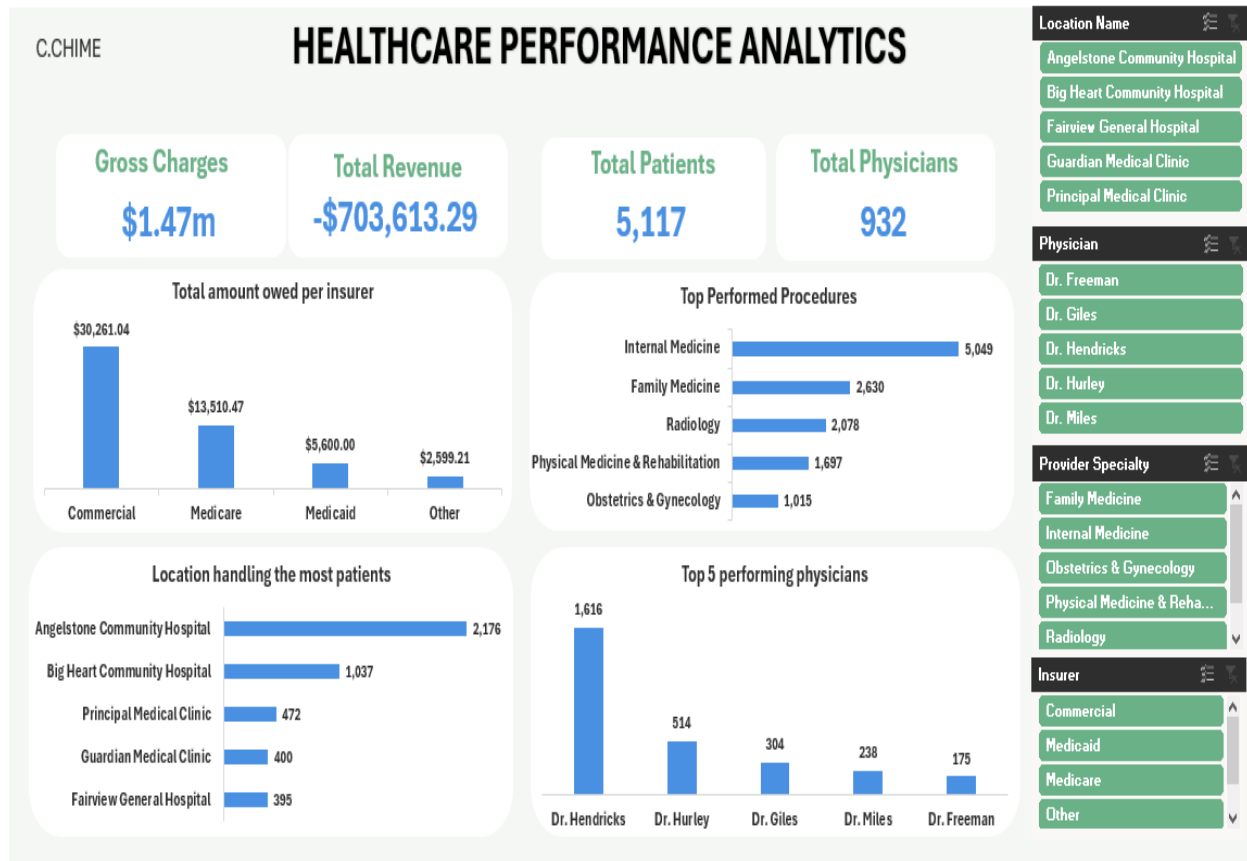


Figure 4: Rank of patients per patients

DASHBOARD



The dashboard reveals several important operational and financial insights within the healthcare organization. Commercial insurance contributes the highest outstanding receivables amount at approximately \$30,251.04, making it the largest source of unpaid balances among insurers. Dr. Hendricks emerges as the highest-performing physician with 1,616 recorded procedures or patient interactions, significantly outperforming other physicians. Internal Medicine is identified as the leading specialty driving hospital operations with 5,049 procedures performed, indicating a strong dependence on general and chronic patient care services. Additionally, Anglestone community hospital handles the highest patient volume with 2176 patients making it the busiest healthcare facility within the network.

The dashboard also highlights a financial concern where the organization generated approximately 1.47 million in gross charges but still recorded a negative revenue value of -703,613.29. This indicates inefficiencies in reimbursements collection, claim processing or revenue cycle management despite strong operational activity.

DISCUSSION

Interpretation of Results

The interpretation of results across the charts shows a consistent pattern of strong operational performance but uneven resource distribution and financial inefficiency. Commercial insurance stands out as the primary contributor to outstanding receivables, indicating challenges in reimbursement and revenue collection. Internal Medicine dominates procedure volume, reflecting its central role in delivering routine and chronic care services, while Anglestone Community hospital handles the highest patient load, suggesting uneven patient distribution across facilities.

At the workforce level, Dr. Hendricks significantly outperforms other physicians in total procedures, highlighting both high productivity and potential workload concentration. Overall, the charts collectively show a healthcare system with high service demand and strong clinical activity, but with financial strain driven by unpaid balances and inefficiencies in revenue cycle management.

Implications

High receivables from commercial insurance will negatively impact cashflow and financial stability. The dominance of Internal Medicine highlights the need for focused resource allocation in that specialty. Heavy reliance on top physicians increases burnout risk and operational vulnerability. The negative revenue despite high charges signals inefficiencies in billing, claims, or reimbursement processes.

CONCLUSION

Summary

This analysis reveals a healthcare system with strong operational performance but weak financial efficiency. Internal medicine and Anglestone Community hospital are the main active drivers, while commercial insurance creates the highest financial burden through outstanding receivables. Dr. Hendricks leads physician performance, reflecting high productivity within the workforce. Overall, the organization requires improved revenue cycle management, better patient distribution and stronger financial controls to align operational success with financial stability

Recommendations

- 1. Strengthening revenue management:** The healthcare organization should strengthen its revenue cycle by improving claims follow-up procedure and implementing stricter billing controls and insurer communication strategies can help reduce outstanding receivables and improve financial stability.
- 2. Workload balance strategies:** Management should introduce workload balancing strategies to prevent overdependence on a few physicians and healthcare facilities. This involves staff redistribution, physician support programs, and improved scheduling systems to ensure operational efficiency.
- 3. Investment in efficient areas:** Additional investment should be made in internal Medicine departments through increased staffing, modern medical equipment, and operational support since the specialty drives the majority of hospital procedures.
- 4. Patient Distribution:** Patients should be distributed across facilities by expanding services in lower-performing hospitals and implementing better referral and awareness systems.